Just as in classical mechanics, once forces are included, the momentum of a particle is not conserved. In fact, Newton's laws of motion can be stated in the form

$$\frac{dp}{dt} = F$$

or

$$\frac{dp}{dt} = -\frac{\partial V}{\partial x}. \qquad (9.12)$$

The rules of quantization require us to add $\mathbf{V}(x)$ to the Hamiltonian,[4]

$$\mathbf{H} = \frac{\mathbf{P}^2}{2m} + \mathbf{V}(x), \qquad (9.13)$$

and modify the Schrödinger equations in the obvious way:

$$\begin{aligned} i\hbar\frac{\partial\psi}{\partial t} &= \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi \\ E\psi &= \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi. \qquad (9.14) \end{aligned}$$

What effect does this have? The additional term certainly affects the way ψ changes with time. That of course must be so if the average position of a wave packet is to follow a classical trajectory. To check our reasoning, let's see if it does. First of all, does Eq. 9.11 still hold? It should, because the connection between momentum and velocity is unaffected by the presence of forces.